

数学物理方程(理科版)

郑神州

北京交通大学数学系

Email: shzhzheng@bjtu.edu.cn

July 11, 2020

第二章 波动方程

§2. 波动方程的初值问题（一维情形）

2.1 问题的简化

一、问题：求解一维波动方程的Cauchy问题

$$\begin{cases} \square u = \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t), & x \in \mathbb{R}, t > 0 \\ u(x, 0) = \varphi(x), & x \in \mathbb{R}, \\ u_t(x, 0) = \psi(x), & x \in \mathbb{R}, \end{cases} \quad (1)$$

简化问题思路：这线性问题，利用叠加原理，将问题分解为三个问题：二个齐次方程和一个非齐次方程构成，问题化为三个相对简单的波动方程Cauchy问题。

$$\begin{cases} \square u_1 = \frac{\partial^2 u_1}{\partial t^2} - a^2 \frac{\partial^2 u_1}{\partial x^2} = 0, & x \in \mathbb{R}, t > 0 \\ u_1(x, 0) = \varphi(x), & x \in \mathbb{R}, \\ u_{1t}(x, 0) = 0, & x \in \mathbb{R}, \end{cases} \quad (2)$$

$$\begin{cases} \square u_2 = \frac{\partial^2 u_2}{\partial t^2} - a^2 \frac{\partial^2 u_2}{\partial x^2} = 0, & x \in \mathbb{R}, t > 0 \\ u_2(x, 0) = 0, & x \in \mathbb{R}, \\ u_{2t}(x, 0) = \psi(x), & x \in \mathbb{R}, \end{cases} \quad (3)$$

$$\begin{cases} \square u_3 = \frac{\partial^2 u_3}{\partial t^2} - a^2 \frac{\partial^2 u_3}{\partial x^2} = f(x, t), & x \in \mathbb{R}, t > 0 \\ u_3(x, 0) = 0, & x \in \mathbb{R}, \\ u_{3t}(x, 0) = 0, & x \in \mathbb{R}, \end{cases} \quad (4)$$

由线性问题叠加原理, 显然有 $u = u_1 + u_2 + u_3$. 下面的定理表明: 只要求出 u_2 , 就可由 u_2 表达式得到 u_1 和 u_3 的表达式.

定理2.1 (齐次化原理) 设 $u_2 = M_\psi(x, t)$ 是定解问题(3)的解, 则定解问题(2)和(4)之解 u_1 和 u_3 表示为

$$u_1(x, t) = \frac{\partial M_\varphi(x, t)}{\partial t}, \quad u_3(x, t) = \int_0^t M_{f_\tau}(x, t - \tau) d\tau,$$

其中 $f_\tau = f(x, \tau)$, 并且假定 $M_\varphi(x, t)$ 和 $M_{f_\tau}(x, t - \tau)$ 分别在区域 $\{(x, t) | x \in \mathbb{R}, t \geq 0\}$ 和 $\{(x, t, \tau) | x \in \mathbb{R}, 0 \leq \tau \leq t < +\infty\}$ 上对变量 x, t 和 τ 充分光滑.

定理2.1的证明: 1. 先证 $u_1 = \frac{\partial}{\partial t} M_\varphi(x, t)$ 满足(2). 由定义 $M_\varphi(x, t)$ 满足:

$$\begin{cases} \square M_\varphi = \frac{\partial^2 M_\varphi}{\partial t^2} - a^2 \frac{\partial^2 M_\varphi}{\partial x^2} = 0, & x \in \mathbb{R}, t > 0 \\ M_\varphi(x, 0) = 0, & x \in \mathbb{R}, \\ (M_\varphi)_t(x, 0) = \varphi(x), & x \in \mathbb{R}, \end{cases}$$

因此

$$\begin{aligned}\square u_1 &= \frac{\partial^2}{\partial t^2} \left(\frac{\partial}{\partial t} M_\varphi(x, t) \right) - a^2 \frac{\partial^2}{\partial x^2} \left(\frac{\partial}{\partial t} M_\varphi(x, t) \right) \\ &= \frac{\partial}{\partial t} \left(\frac{\partial^2}{\partial t^2} M_\varphi(x, t) - a^2 \frac{\partial^2}{\partial x^2} M_\varphi(x, t) \right) = 0.\end{aligned}$$

$$u_1(x, 0) = \left[\frac{\partial}{\partial t} M_\varphi(x, t) \right] \Big|_{t=0} = \varphi(x),$$

$$\frac{\partial u_1}{\partial t} \Big|_{t=0} = \frac{\partial^2 M_\varphi(x, t)}{\partial t^2} \Big|_{t=0} = a^2 \frac{\partial^2 M_\varphi(x, t)}{\partial x^2} \Big|_{t=0} = a^2 \frac{\partial^2 M_\varphi(x, 0)}{\partial x^2} = 0.$$

因此 $u_1 = \frac{\partial}{\partial t} M_\varphi(x, t)$ 满足问题(2).

2. 再证 $u_3(x, t) = \int_0^t M_{f_\tau}(x, t - \tau) d\tau$ 满足(4). 注意到 $M_{f_\tau}(x, t)$ ($\tau \geq 0$) 满足:

$$\begin{cases} \square M_{f_\tau}(x, t) = \frac{\partial^2 M_{f_\tau}}{\partial t^2} - a^2 \frac{\partial^2 M_{f_\tau}}{\partial x^2} = 0, & x \in \mathbb{R}, t > 0 \\ M_{f_\tau}(x, 0) = 0, & x \in \mathbb{R}, \\ [(M_{f_\tau})(x, t)]|_{t=0} = f_\tau(x) = f(x, \tau), & x \in \mathbb{R}, \end{cases}$$

于是

$$u_3(x, t)|_{t=0} = \left(\int_0^t M_{f_\tau}(x, t - \tau) d\tau \right) \Big|_{t=0} = 0,$$

$$\begin{aligned} u_3(x, t)_t|_{t=0} &= \left((M_{f_\tau}(x, t - \tau))|_{\tau=t} + \int_0^t \frac{\partial}{\partial t} M_{f_\tau}(x, t - \tau) d\tau \right) \Big|_{t=0} \\ &= \left((M_{f_\tau}(x, 0)) + \int_0^t \frac{\partial}{\partial t} M_{f_\tau}(x, t - \tau) d\tau \right) \Big|_{t=0} = 0, \end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 u_3(x, t)}{\partial t^2} &= \frac{\partial}{\partial t} \left[M_{f_t}(x, 0) + \int_0^t \frac{\partial}{\partial t} M_{f_\tau}(x, t - \tau) d\tau \right] \\
&= \left(\frac{\partial}{\partial t} M_{f_\tau}(x, t - \tau) \right) \Big|_{\tau=t} + \int_0^t \frac{\partial^2}{\partial t^2} M_{f_\tau}(x, t - \tau) d\tau \\
&= \left(\frac{\partial}{\partial t} M_{f_\tau}(x, t - \tau) \right) \Big|_{\tau=t} + \int_0^t a^2 \frac{\partial^2}{\partial x^2} M_{f_\tau}(x, t - \tau) d\tau \\
&= \left(\frac{\partial}{\partial t} M_{f_\tau}(x, t - \tau) \right) \Big|_{\tau=t} + a^2 \frac{\partial^2}{\partial x^2} \int_0^t M_{f_\tau}(x, t - \tau) d\tau \\
&= \left(\frac{\partial}{\partial t} M_{f_\tau}(x, t - \tau) \right) \Big|_{\tau=t} + a^2 \frac{\partial^2 u_3(x, t)}{\partial x^2}.
\end{aligned}$$

记: $\Phi(x, t, \tau) = M_{f_\tau}(x, t)$, 则

$$\frac{\partial}{\partial t} M_{f_\tau}(x, t - \tau) = \frac{\partial}{\partial t} \Phi(x, t - \tau, \tau) = \frac{\partial}{\partial t} \Phi(x, t, \tau)|_{t=t-\tau}$$

所以

$$\begin{aligned} \left(\frac{\partial}{\partial t} M_{f_\tau}(x, t - \tau) \right) |_{\tau=t} &= \left[\frac{\partial}{\partial t} \Phi(x, t, \tau) |_{t=t-\tau} \right] |_{\tau=t} \\ &= \left[\frac{\partial}{\partial t} \Phi(x, t, \tau) |_{t=0} \right] |_{\tau=t} \\ &= \left\{ \left[\frac{\partial}{\partial t} M_{f_\tau}(x, t) |_{t=0} \right] \right\} |_{\tau=t} \\ &= f_\tau(x) |_{\tau=t} = f(x, \tau) |_{\tau=t} \\ &= f(x, t). \end{aligned}$$

移项，于是

$$\frac{\partial^2 u_3(x, t)}{\partial t^2} = f(x, t) + a^2 \frac{\partial^2 u_3(x, t)}{\partial x^2}$$

即

$$\frac{\partial^2 u_3(x, t)}{\partial t^2} - a^2 \frac{\partial^2 u_3(x, t)}{\partial x^2} = f(x, t).$$

证毕.

附注1: 在定理证明中, 没考虑各运算合法性。因此这一公式只是一种形式上的公式。偏微分方程的求解过程往往采用这样的步骤: 先求形式解, 即在求解中不妨先假定所有的运算都合法的, 然后再讨论对定解资料要加些什么条件才能保证所得的形式解确实是真正的解, 即具有所需要的连续可微性。

附注2: 本定理也适用于高维的波动方程Cauchy问题。

附注3: 对于热传导方程的Cauchy问题也有类似齐次化原理。对于带有齐次边界条件的热传导方程和波动方程的混合定解问题, 也有类似于本定理的结论。

附注4: 对于常微分方程的初值问题情形, 也有类似于本定理的结论。

同学们，再见！